MIL-B-23272B
19 June 1970
SUPERSEDING
MIL-B-0023272A
29 May 1967
MIL-B-23272
26 April 1962

MILITARY SPECIFICATION

BATTERY, STORAGE:
ALKALINE, NICKEL-CADMIUM

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for nickel-cadmium storage batteries for use in military vehicles (see 6.1).

1.2 Classification. Batteries shall be of the following types, as specified (see 6.2):

Type 2HNC	12-volt, 35-ampere-hour capacity.
Type 6TNC	12-volt, 70-ampere-hour capacity.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS
Military
MIL-P-514 - Plates, Identification, Transportation, Instruction and Marking, Blank.
MIL-B-23272/1 - Battery, Storage: Alkaline, Nickel-Cadmium, BB-634/u ( ), 12-volt, 70 Amp Hr. Capacity.

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.

FSC-5140

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:16Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The battery furnished under this specification shall be a product which has been tested, has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list (see 4.2 and 6.3).

3.2 Materials. Material shall be as specified herein or on drawings, standards or applicable military specification sheet and shall be free of defects which would adversely affect performance.

3.2.1 Cell sealing compound. The cell sealing compound shall be alkali resistant, shall not shrink or separate from adjoining surfaces, and shall maintain an unbroken seal throughout an ambient temperature range of minus 65°F to plus 200°F. The individual cells shall be sealed to maintain the seal when subjected to a pressure of two pounds per square inch (psi) throughout the ambient temperature range of minus 65°F to plus 200°F.

3.2.2 Electrolyte. The electrolyte used shall be a solution of potassium hydroxide and deionized or distilled water with a specific gravity of 1.300 ± .020 at plus 72°F. The electrolyte shall be furnished in the battery.

3.3 Design and construction. Unless otherwise specified on the applicable drawing or specification sheet (see MIL-B-23272/1), the battery shall be designed and constructed to the form and dimensions specified herein (see figures 1 and 2).

3.3.1 Containers.

3.3.1.1 Battery container. The battery container shall be of one piece, monobloc design. The container material shall be of a high strength alkali resistant non-porous plastic which is resistant to temperatures ranging from minus 65°F to plus 200°F. The battery container shall conform to figure 1, figure 2 or the applicable specification sheet.

3.3.2 Cover. The cover shall be one piece high strength alkali resistant non-porous plastic. It shall insulate and protect all intercell connections enabling the battery to be submersible. There shall be no exposed intercell connections. The only exposed parts carrying current shall be the positive and negative terminal posts. The cover shall conform to figure 4 and figure 1, figure 2, or the applicable specification sheet.

Source: https://assist.dla.mil -- Downloaded: 20261 0-19T16:16Z
Check the source to verify that this is the current version before use.

3.3.3 Intercell connectors. Unless otherwise specified (see 6.2), intercell connectors and other current carrying parts shall have sufficient cross sectional area to withstand the charge or discharge rate requirements for each specific battery type to which they pertain.

3.3.4 Terminal posts. Terminal posts shall conform to figure 3.

3.3.5 Corrugation. The sides of the external case shall be corrugated to conform to figure 4.

3.3.6 Handles. Unless otherwise specified (see 6.2), handles shall be provided and shall be so designed to support the battery weight during lifting throughout a temperature range of minus 65°F to plus 200°F. Handles shall conform to figure 5.

3.3.6.1 Metal handles. Metal handles shall be of material that either by itself or when otherwise treated shall resist electrolyte KOH solution of 1.300 specific gravity, and when tested in accordance with 4.7.2.2 shall show no corrosion or alkali absorption that interferes with performance of its function. The handles shall operate freely in their retainers. Metal handles shall conform to figure 5.

3.3.6.2 Plastic handles. Plastic handles shall be rope type. When tested as specified in 4.6.2.3, plastic rope handles and bonds shall evidence no ruptures, cracks, distortions, or other damage. Plastic handles shall conform to figure 5.

3.3.7 Vent plugs. Vent plugs shall be furnished for each cell, and shall be located as shown in figure 1, figure 2, or the applicable specification sheet. Materials for vent plugs shall be non-absorbent and alkaline resistant. Plugs must be removable by hand without the use of any tool. Vent plugs shall have a gas or pressure release mechanism that is not affected or made inoperative by carbonate deposits. The vent plugs and their release or venting device must be operative throughout a temperature range of minus 65°F to plus 250°F.

3.4 Performance.

3.4.1 Assembled battery.

3.4.1.1 Ampere-hour capacity. When tested as specified in 4.5.3.1, the battery shall have an ampere-hour capacity not less than that specified in table I.

3.4.1.2 Cranking ability. When tested as specified in 4.5.3.2, the battery shall deliver the specified current for not less than the period of time

Source: https://assist.dla.mil -- Downloaded: 20261O-19T16:16Z
Check the source to verify that this is the current version before use.

specified in table I. During the first five seconds of the test, the terminal voltage shall drop no lower than the values specified in table I.

Table I. Electrical performance characteristics

Battery type	2HNC	6TNC
Number of cells	10	10
Weight filled	40 lbs	80 lbs
Nominal voltage	12 volts	12 volts
Rated capacity 5-hour rate	35 amp-hr	70 amp-hr
Cranking ability:		
Discharge rate	150 amps	300 amps
Minimum time at minus 40°F	5.5 minutes	5.5 minutes
Minimum time at plus 80°F	10.0 minutes	10.0 minutes
Minimum 5-second voltage at minus 40°F	8.0 volts	8.0 volts
Life cycle minimum capacity	250	300
Overcharge minimum capacity	9 cycles	14 cycles

3.4.1.3 Endurance. When tested as specified in 4.5.3.3, the battery shall complete not less than the number of cycles specified in table I.

3.4.1.4 Overcharge cycle capacity. When tested as specified in 4.5.3.4, the battery shall complete not less than the number of overcharge cycles specified in table I.

3.4.1.5 Retention of charge. When tested as specified in 4.5.3.5, sample three battery (see table II) shall show no more than 20 percent decrease in rated five-hour capacity at the plus 80°F ± 5°F temperature capacity discharge and sample four battery shall show not more than 80 percent decrease at the high rate as specified in table I at the minus 40°F temperature capacity discharge.

3.4.1.6 Temperature rise. When tested as specified in 4.5.3.6, the battery shall have ceased to show a temperature rise at the end of the two-hour charging period, and the battery shall not have sprayed or lost any of its electrolyte throughout the charge. The battery shall then satisfactorily meet 80 percent of the rated ampere-hour capacity as specified in table I.

3.4.2 Battery components.

Source: https://assist.dla.mil -- Downloaded: 202610-19T16:16Z
Check the source to verify that this is the current version before use.

3.4.2.1 Dielectric strength. When tested as specified in 4.5.4.1, the container shall be free of leaks and other imperfections, and shall evidence no perforation or burning through.

3.5 Physical.

3.5.1 Assembled battery.

3.5.1.1 Vibration resistance. When tested as specified in 4.6.1.1, the battery shall maintain a steady voltage and current with no loosening of any component parts of the connectors, flooding of electrolyte at the top, broken connections, straps, plates or other damage which would result in a reduction of required performance.

3.5.2.1 Vent plug leakage. When tested as specified in 4.6.2.1, the vent plugs shall not leak at a rate in excess of two drops of water in 10 seconds, and shall open to permit discharge of gas before the internal pressure reaches a maximum of 0.5 pounds per square inch gage (psig) in excess of the external pressure. Test samples shall be selected from batteries that have completed the endurance test specified in 4.5.3.3.

3.5.2.2 Terminal post torque resistance. When tested as specified in 4.6.2.2, the terminal posts shall withstand a torque of 250 pound-inches without causing distortion, leakage, or other damage to the battery.

3.5.2.3 Dielectric strength. When tested as specified in 4.5.4.1, the battery container shall be free of leaks and other imperfections, and shall evidence no perforation, burning through or breakdown of insulation.

3.6 Environmental.

3.6.1 Assembled battery.

3.6.1.1 Thermal shock resistance. When tested as specified in 4.7.1.1, the battery shall evidence no leakage or other damage which would result in a reduction of performance.

3.6.2 Battery components.

3.6.2.1 Alkali absorption resistance. When tested as specified in 4.7.2.1, the cell container material shall increase a maximum of 1.5 percent in weight, a maximum of two percent in any dimension, and shall show no cracks or blisters.

3.6.2.2 Corrosion resistance. When tested as specified in 4.7.2.2, metal parts shall show no evidence of corrosion affecting the function of these parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:16Z
Check the source to verify that this is the current version before use.

3.7 Weight. The gross weight of the battery shall not exceed that specified in table I.

3.8 Marking. The battery shall be marked in accordance with the equipment marking requirements of MIL-STD-130 and as specified on the applicable drawing. When specified on the applicable drawing, nameplate shall be in accordance with MIL-P-514. Marking shall include the following information in the order shown:

Battery, Nickel Cadmium.
Type.
Rated voltage.
Rated capacity.
Manufacturer's serial number.
Manufacturer's name.
Date of manufacture.
Contract or order number.
U.S. Government Property.

3.8.1 Polarity marking. The polarity of the terminal posts shall be marked for identification. In lieu of the "POS" and "NEG", the positive terminal post may be identified by a "P" or a " + " sign, and the negative terminal post may be identified by an "N" or " - " sign. Marking shall be permanent and non-deteriorating.

3.8.2 Warning marking. The battery shall be furnished with a warning plate located near the filler plugs specifying the use of alkaline electrolyte only and specifying the proper fill level of electrolyte. Marking shall be permanent and non-deteriorating.

3.9 Instructions. Instruction for placing the battery into service shall be provided with each battery. The instructions shall be printed on a plate attached to each battery container on the side opposite the nameplate. Also, a service manual shall be furnished with each battery providing the following information: Instructions for placing the battery into service; care and maintenance instructions; charging and discharging characteristics and instructions; charging methods, rates and voltages; electrolyte contamination and safety precautions.

3.10 Workmanship. Workmanship shall be such as to assure a product free of imperfect seals and bonds. Metallic parts shall be free of burs, rust, scratches, chips, sharp edges, or other defects which will affect serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all

Source: https://assist.dla.mil -- Downloaded: 20260-19T16:16Z
Check the source to verify that this is the current version before use.

inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests. A qualification sample of five batteries of each type to be tested shall be submitted for qualification testing. The samples shall be representative of the units proposed to be furnished to the Government. Qualification testing shall be conducted by the Government at a place designated by the Government, and shall consist of examination for the defects specified in 4.3.2.2, and testing as specified in table II in the order listed.

Table II. Order of qualification testing

Sample	Paragraph	Test
1 and 2	4.5.3.1	Ampere-hour capacity for fully charged batteries
	4.5.3.2	Cranking ability at plus 80°F
	4.5.3.2	Cranking ability at minus 40°F
	4.5.3.3	Endurance
3 and 4	4.5.3.5	Retention of charge
	4.5.3.6	Temperature rise
	4.7.1.1	Thermal shock resistance
	4.5.3.4	Overcharge cycle capacity
5	4.6.1.1	Vibration resistance
	4.5.4.1	Dielectric strength
	4.6.2.2	Terminal post torque resistance
	4.6.2.1	Vent plug leakage
	4.7.2.2	Corrosion resistance
Cell container	4.7.2.1	Alkali absorption resistance

4.2.1 Failure. Failure of a qualification sample to pass any of the examinations or tests specified herein may be cause for the Government to refuse to conduct additional testing until the faults revealed by the tests have been corrected.

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 Lot formation. Unless otherwise specified (see 6.2), a lot shall consist of all batteries of one type, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:16Z
Check the source to verify that this is the current version before use.

4.3.1.2 Sampling for quality conformance examination. Samples for acceptance examination shall be selected in accordance with MIL-STD-105.

4.3.1.3 Sampling for quality conformance testing. Samples for acceptance testing shall be selected in accordance with inspection level S-3 of MIL-STD-105.

4.3.2 Examination.

4.3.2.1 Acceptable quality level. Each sample selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.3.2.2 Classification of defects. For examination purposes defects shall be classified in table III.

Table III. Classification of defects for batteries in accordance with MIL-STD-105

Categories	Defects	Method of Inspection
Critical:		
Major		
102	Dimensions affecting interchangeability not within tolerance (see 3.3).	Gage
101	Nonconformance of required materials (see 3.2).	Visual
103	Nonconformance in design and construction (see 3.3).	Visual
104	Improper marking (see 3.8).	Visual
105	Missing instructions (see 3.9).	Visual
Minor		
201	Dimensions not affecting interchangeability not within tolerance (see 3.3).	Visual
202	Faulty workmanship (see 3.10).	Gage

4.3.3 Testing.

4.3.3.1 Quality conformance tests. Samples selected in accordance with 4.3.1.3 shall be subjected to the tests specified in table IV in the order specified using an AQL of 6.5 on the basis of percent defective.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:16Z
Check the source to verify that this is the current version before use.

Table IV. Order of acceptance testing

Paragraph	Test
4.5.3.1	Ampere-hour capacity
4.5.3.2	Cranking ability
4.6.2.2	Vent plug leakage
4.6.2.1	Terminal post torque
4.6.1.1	Vibration

4.4 Control tests.

4.4.1 Sampling for control tests. Samples for control testing shall be selected at the rate of two of each 50 batteries produced, except that no less than two nor more than six batteries shall be selected in any 30 day period. Batteries shall be examined for the defects specified in 4.3.2.2, and subsequently subjected to the tests specified in table VI in the order listed.

Table VI. Order of control testing

Sample	Paragraph	Test
1 and 2	4.5.3.1	Ampere-hour capacity
	4.5.3.2	Cranking ability
	4.6.2.1	Vent plug leakage
	4.6.1.1	Vibration resistance
	4.7.1.1	Thermal shock resistance

4.4.2 Failure. Failure of a control test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the examination or test have been corrected.

4.5 Performance tests.

4.5.1 General.

4.5.1.1 Standard test conditions. Tests shall be conducted under general laboratory conditions acceptable to the Government. Unless otherwise specified herein, all tests shall be conducted at ambient room air temperature, ambient atmospheric pressure and ambient relative humidity. The ambient room temperature shall be maintained at 77° ± 15°F. Unless otherwise specified herein, test specimens shall be thermally stabilized for 24 hours prior to being subjected to tests.

4.5.1.2 Temperature of electrolyte. Unless otherwise specified herein, the temperature of the electrolyte at the beginning of the test shall be within ± 2°F of the test temperature.

Source: https://assist.dla.mil -- Downloaded: 20260-19T16:16Z
Check the source to verify that this is the current version before use.

4.5.1.3 Apparatus. In addition to standard measuring equipment, test apparatus shall include chambers for conducting low temperature (-65°F) and high temperature (≠ 200°F) tests and a submersion tank.

4.5.2 Instrument accuracy.

4.5.2.1 Calibration of test equipment. Test equipment accuracy shall be such as to permit measurement in ten percent of product or test specification tolerance. Records of calibration of test equipment at intervals sufficient to establish required accuracy shall be made available to the Government. The inspector may refuse to conduct an inspection where accuracy of test equipment has not been established to the satisfaction of the Government.

4.5.2.2 Hydrometer. The hydrometer shall provide a float accuracy of plus or minus three points (.003) specific gravity scale and temperature range.

4.5.2.3 Electrical indicating instruments. All voltmeters and ammeters used in testing the batteries shall be accurate within ≠ 0.5 percent of the full-scale value. The voltmeter and ammeter ranges shall be such that all readings are taken on the upper half of the scale.

4.5.3 Assembled battery.

4.5.3.1 Ampere-hour capacity test. The battery shall be charged at a constant potential equivalent to 1.55 volts per cell until such time as three consecutive current readings taken at half hour intervals show no decrease in current in excess of one ampere. After a minimum stand-time of two hours, the battery, stabilized and maintained at 80° ≠ 5°F shall be discharged at the five hour rate (that current in amperes equal to 1/5 of the battery's rated ampere-hour capacity) to a final terminal voltage equivalent to an average of 1.05 volts per cell. To determine conformance to 3.4.1.1, the time required to accomplish this discharge shall be recorded and results used to calculate the ampere-hour capacity of each battery and the average capacity of the batteries tested. The test shall be repeated two additional times. Each battery shall yield an average capacity (ampere-hours) of not less than that specified.

4.5.3.2 Cranking ability test. After testing as specified in 4.5.3.1 and while stabilized at the ambient temperature of 80° ≠ 5°F the battery shall be charged at a constant potential equivalent to 1.55 volts per cell until such time as three consecutive current readings taken at half hour intervals show no decrease in current in excess of one ampere. The battery shall then be placed in a cold chamber at an ambient temperature of minus 40° ≠ 1°F for a period of 24 hours. While maintaining the ambient temperature of minus 40° ≠ 1°F, the battery shall be discharged at the discharge rate as specified in table I for the applicable battery type, until the battery terminal voltage is equivalent to 0.6 volt per cell. For the 80 ≠ 5°F cranking ability test, the battery shall be recharged as indicated above. The cranking ability test shall be repeated at 80

Source: https://assist.dla.mil -- Downloaded: 202610-19T16:16Z
Check the source to verify that this is the current version before use.

± 5°F. This test may be repeated the second time should the batteries fail the first test. The 80°F and low temperature test shall be conducted aternately. To determine conformance to 3.4.1.2, the battery voltage after five seconds of discharge and the total time of discharge shall be recorded. Failure of a test sample to pass this test shall be cause for the sample to be retested. The retest battery shall be tested as previously specified. Failure of the retest battery to pass this second cycle shall be considered as failure to meet the specified requirements.

4.5.3.3 Endurance test. The battery shall be subjected to normal and conditioning cycles as specified herein. Complete discharging and shorting of the battery to balance the individual cell capabilities may be conducted when required by poor battery performance.

4.5.3.3.1 Normal cycles. After having been given the capacity discharge specified in 4.5.3.1, the battery shall be charged at the five-hour rate until three recorded charging voltage readings, taken one hour apart, show no increase in value. Normal and conditioning cycles shall be made with the battery in a water bath maintained at a temperature of 100° ± 5°F. Water shall be added when necessary but not more often than every other day and only when the battery is in a charged condition. The test shall consist of a series of six-hour normal cycles (four cycles per day, or approximately 24 per week). Each cycle shall consist of discharge for one hour at 50 percent of rated capacity and charged for five hours at 14 percent of its rated capacity.

4.5.3.3.2 Weekly capacity discharge test cycles. During the normal cycle portion of the endurance test, in order to check the condition of the battery, the ampere-hour capacity shall be determined at the completion of each 24 normal cycles. The battery shall be discharged at the ampere rate for normal cycles until the terminal voltage reaches an equivalent of an average of 1.05 volts per cell. The ampere-hour capacity shall be calculated as the product of the current rate in amperes and the time of discharge in hours. Following this discharge, the battery shall be fully charged at the five-hour rate and the normal cycle procedure shall be continued. The battery shall be placed on a discharge cycle which, with the charge cycle, shall constitute a full normal cycle (see 4.5.3.3.1).

4.5.3.3.3 Completion of test. During the normal cycling or weekly capacity discharge test cycles, if the ampere-hour capacity equals or drops below 50 percent of the rated five-hour capacity; the battery shall be fully charged at the five-hour rate and tested as specified in 4.5.3.3.2. If the capacity equals or falls below 50 percent of rated ampere-hour (AH) capacity, the endurance test shall be considered concluded and if above 50 percent the endurance test shall be continued. However, if the capacity drops below 50 percent of the rated AH capacity before an additional 24 normal cycles have been completed; the endurance test shall be concluded. The cycles attained by the battery shall be taken as the total of the actual normal cycles, plus each cycle received on other tests. After weekly capacity discharge tests and after completion of the life cycle

Source: https://assist.dla.mil -- Downloaded: 202610-19T16:16Z
Check the source to verify that this is the current version before use.

requirement, but before any maintenance (watering and cleaning) the vent plugs shall be tested to determine compliance to the internal pressure release requirement (see 4.6.2.1). Battery cells shall also be tested at this time for compliance to part (a) of the thermal shock requirement (see 4.7.1.1). The battery external container and handles shall be functional at the end of the test.

4.5.3.4 Overcharge cycle capacity. The test shall be conducted with the sample in a water bath maintained at a temperature of 100° ± 5°F. Water shall be added to the battery as needed to replace evaporation except that no water shall be added while on open circuit phase. The battery, initially discharged, shall be subjected to overcharge cycles as specified in 4.5.3.4.1, 4.5.3.4.2 and 4.5.3.4.3.

4.5.3.4.1 Charge phase. A continuous charge in ampere-hours equal to ten times the normal rated ampere-hour capacity shall be applied at a current equal to the rated capacity divided by ten.

4.5.3.4.2 Open circuit phase. The battery shall be allowed to stand on open circuit for a minimum of 48 hours.

4.5.3.4.3 Dishcharge phase. The battery, while still at the temperature specified in 4.5.3.4, shall be discharged at the high rate specified in table I to a terminal voltage equivalent to an average of 0.6 volt per cell. The length of time required for such discharge shall be recorded. With no charging of the battery other than as specified in 4.5.3.4.1, the cycle shall be repeated until discharge time is 30 seconds or less or the number of cycles specified in table I has been reached. The number of cycles and discharge rate shall be recorded to determine conformance to 3.4.1.4.

4.5.3.5 Retention of charge test. Prior to testing, the batteries shall be charged and discharged as specified in 4.5.3.1. This charge and discharge shall be conducted for three cycles. The fully charged battery shall be allowed to stand on open circuit for 30 days at an ambient temperature of 80° ± 10°F after which sample three (see table II) shall be discharged at the five-hour rate, and sample four shall be discharged at the high rate as specified in table I while at minus 40°F to determine conformance to 3.4.1.5.

4.5.3.6 Temperature rise test. The battery, while fully charged, shall be placed in an ambient temperature of 120° ± 5°F for a period of 24 hours. The battery shall then be discharged at the high rate specified in table I to a terminal voltage equivalent to an average of 0.6 volt per cell. Immediately following this discharge, the battery shall be placed across a potential of 15 volts and charged for a period of two hours and then discharged at the five-hour rate to determine conformance to 3.4.1.6. After successful testing, the temperature shall be lowered to 80° ± 5°F and the battery shall be subjected to the test specified in 4.5.3.1 to determine conformance to 3.4.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:16Z
Check the source to verify that this is the current version before use.

4.5.4 Battery components.

4.5.4.1 Dielectric strength test. To determine conformance to 3.4.2.1, the battery with all external surfaces and parts clean and dry, shall be subjected to a test voltage of 2,500 volts root mean square (rms) at a frequency of 60 cycles per second (cps) between current-carrying parts and the container for 60 seconds. A breakdown of insulation during this test shall constitute a failure.

4.6 Physical tests.

4.6.1 Assembled battery.

4.6.1.1 Vibration resistance test. The battery, while in a fully-charged state and at an ambieg temperature of 80° ± 10°F, shall be subjected to a vibration consisting of simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum total excursion). The frequency shall be varied uniformly between the approximate limits of ten and 55 cps and return to ten cps and shall be traversed in approximately one minute. The test shall be performed for one hour in each of the three mutual perpendicular planes (for a total period of three hours). During the test, the battery shall be discharged at the five-hour rate. After testing a specified temperature, the battery shall be placed in a cold box and stabilized at a temperature of minus 40° ± 1°F. Immediately after removal from the cold box, the test shall be repeated for twenty minutes in each of the three mutual perpendicular planes (for a total period of one hour). During the tests, the battery shall be observed for maintenance of steady voltage and current. After testing, the battery shall be examined to determine conformance to 3.5.1.1. The vibration test shall be followed immediately by dielectric strength test, 4.5.4.1 and part "a" of 4.7.1.1, thermal shock resistance test within 24 hours.

4.6.2 Battery components.

4.6.2.1 Vent plug leakage test. To determine conformance to 3.5.2.1, each vent plug shall be inserted in a fixture arranged so that water pressure can be applied on the upper side of the plug, and an air pressure on the lower. Using this fixture, the upper side of each plug shall be subjected to a water pressure which exceeds the air pressure on the lower side by two psi. The number of drops of water that leak through each vent plug and the time required for such leakage shall be recorded for a period not less than one minute. The water pressure shall then be released and the air pressure on the lower side increased to 0.5 psi (or until the vent opens) above the pressure existing on the upper side.

4.6.2.2 Terminal post torque resistance test. An increasing torque up to 250 pound-inches shall be applied in a direction perpendicular to the axis of the terminal posts by a torque indicating wrench or other torque indicating device, through a fitted battery terminal or other clamping device. Watch shall be kept for any evidence of distortion of the strap and connected plates, or of

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:16Z
Check the source to verify that this is the current version before use.

the seal between the posts and the cell cover. If no evidence of distortion or visible damage to post seal is revealed on waterproof batteries, two psi internal pressure shall be applied for two minutes to each cell tested for resistance to torque. No drop in internal pressure shall take place to determine conformance to 3.5.2.2.

4.6.2.3 Plastic rope handle test. The handle and the bond areas shall be saturated with electrolyte conforming with 3.2.2 and air dried twice daily for two days. The handles and bond areas shall again be saturated. The battery shall then be placed in an ambient air temperature of plus 190° ± 2°F for 60 minutes, allowed to cool to room temperature, and again heated at 190° ± 2°F for 60 minutes. The battery shall be removed from the oven and immediately lifted by each handle in a vertical direction with an accelerating force of two times the filled battery weight, but not to exceed 200 pounds. The supporting rope shall be taut prior to the application of the accelerating force. The handles shall be supported over a length of 4 1/2 inches during test. The handle support shall have a radius of 4 1/2 inches. The lift shall be repeated three times. The battery shall clear the floor or stand on each lift. At the conclusion of the three lifts, the handles and bond shall be examined for conformance to 3.3.6.2. After the test, the battery shall be placed in an ambient air temperature of minus 65° ± 2°F for 24 hours. The battery shall be removed from the cold box and immediately subjected to three lifts as previously specified. At the conclusion of the three lifts, the handles and bond shall be examined for conformance to 3.3.6.2.

4.7 Environmental tests.

4.7.1 Assembled battery.

4.7.1.1 Thermal shock resistance test. The battery shall be subjected to the following cycles to determine conformance to 3.6.1.1. An internal pressure of two psi shall be applied individually to each cell in parallel with a manometer immediately upon removal from the cold chamber or oven, as applicable. Drop in pressure in the individual cells shall be regarded as indication of leakage. The following procedure shall be used:

(a) A two psi internal pressure shall be applied for two minutes with the battery stabilized at an ambient temperature of plus 80° ± 5°F.
(b) The pressure shall be removed. The battery shall be placed in an ambient temperature of minus 65° ± 2°F for 24 hours or until the battery is stabilized then subjected to a two psi internal pressure for two minutes.
(c) The pressure shall be removed and the battery shall immediately be placed in an ambient temperature of plus 190° ± 5°F for 24 hours and then subjected to a two psi internal pressure for two minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:16Z
Check the source to verify that this is the current version before use.

(d) Cycle in accordance with subparagraph b shall then be repeated immediately.
(e) Cycle in accordance with subparagraph c shall then be repeated immediately.

4.7.2 Battery components.

4.7.2.1 Alkali absorption resistance test. Two specimens, each three by three inches, shall be cut from the battery container. The cut edges shall be neither polished nor sealed. After being measured with calipers and weighed in the dry condition at an ambient temperature of 80° ± 10°F, each specimen shall be immersed in a covered vessel containing 150cc of alkaline solution of 1.300 specific gravity at 72°F. This vessel shall be held for seven days in an oven at 150° ± 4°F. At the end of the heating period, the specimens shall be rinsed in water and dried on the surface. They shall then be inspected for evidence of cracks or blisters and shall be measured and weighed, and the percentage increase in dimensions and weight shall be calculated to determine conformance to 3.3.6.1.

4.7.2.2 Corrosion resistance test. All metal parts shall be subjected to the salt spray test specified in Federal Test Method Standard 151 for a continuous period of 200 hours. Thereafter they shall also be immersed for 48 hours in alkaline solution of 1.300 specific gravity (at 72°F). At the conclusion of each test, all specimens shall be examined for conformance to 3.3.6.1.

4.8 Inspection of preparation for delivery.

4.8.1 Materials and processes. All materials and processes involved in the preparation for delivery, shall be inspected to determine conformance to requirements of section 5. Any evidence of deviation from specified requirements shall be cause, at the option of the Government, for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection, shall be in accordance with the applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The battery covered by this specification is intended primarily for service in military vehicle, starting of missile support ground power equipment, radio operation and as a source of electrical energy for operating vehicular accessories control mechanisms.

Source: https://assist.dla.mil -- Downloaded: 202610-19T16:16Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of battery required (Military Specification Sheet and FSN Number).
(c) Any special intercell connector (see 3.3.3).
(d) Any special handles (see 3.3.6).
(e) Any special lot size (see 4.3.1.1).
(f) Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the U. S. Army Tank-Automotive Command, Warren, Michigan 48090 and information pertaining to qualification of products may be obtained from that activity.

Custodians:
Army - AT
Air Force - 82

User activities:
Army - EL, ME

Preparing activity:
Army - AT
Project No. 6140-0198

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:16Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:16Z
Check the source to verify that this is the current version before use.

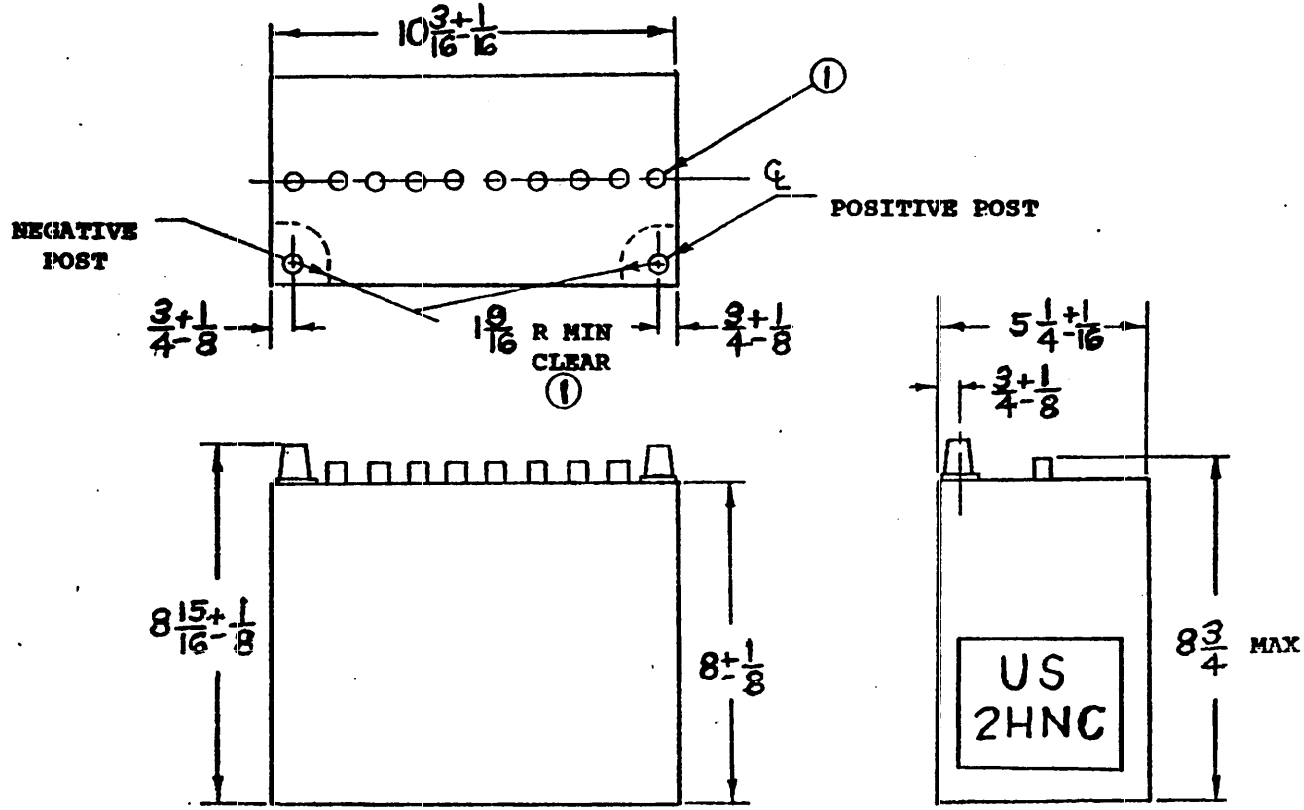


Figure 1 -BATTERY, STORAGE, NICKEL-CADMIUM

BATTERY TYPE	NUMBER OF CELLS	NOMINAL VOLTAGE	AMP HOUR CAPACITY 5 HR RATE	CRANKING ABILITY			LIFE CYCLE	OVERCHARGE
				MINIMUM TIME MINUTES		MINIMUM 5 SECOND VOLTS	MINIMUM CYCLE	MINIMUM CYCLE
				DISCHARGE RATE		150 AMPS		
				-40°F	+80°F	-40°F		
2HNC	10	12	35	5.5	10.0	8.0	250	9

NOTES:

1. VENT PLUGS MUST BE REMOVABLE WITHOUT USE OF ANY TOOL OR DEVICE. PLUG LOCATIONS SHALL BE FIXED ONLY IN RESPECT TO REQUIRED MINIMUM DISTANCE FROM TERMINAL POSTS AND SIDES OF BATTERY.
2. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.

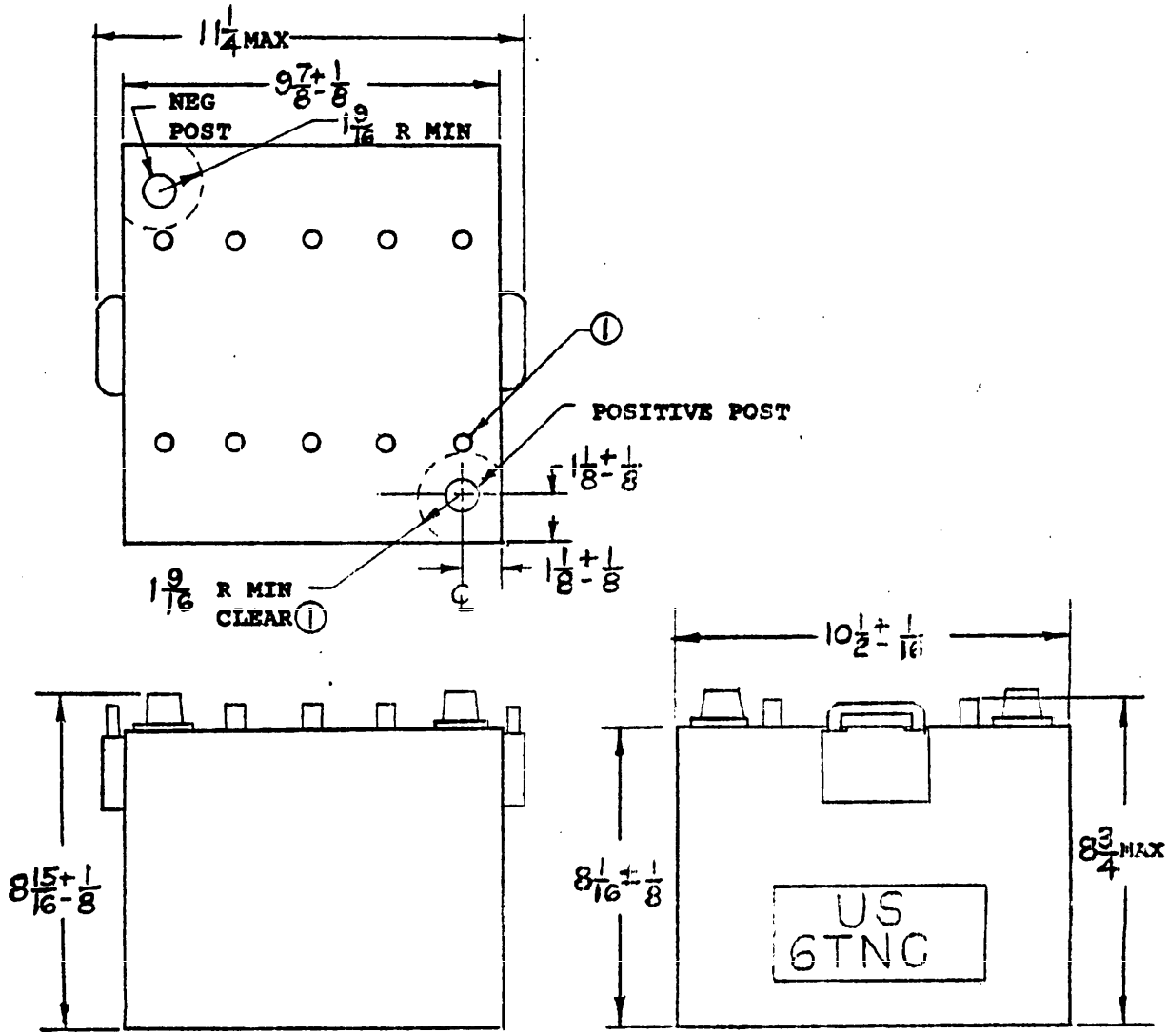


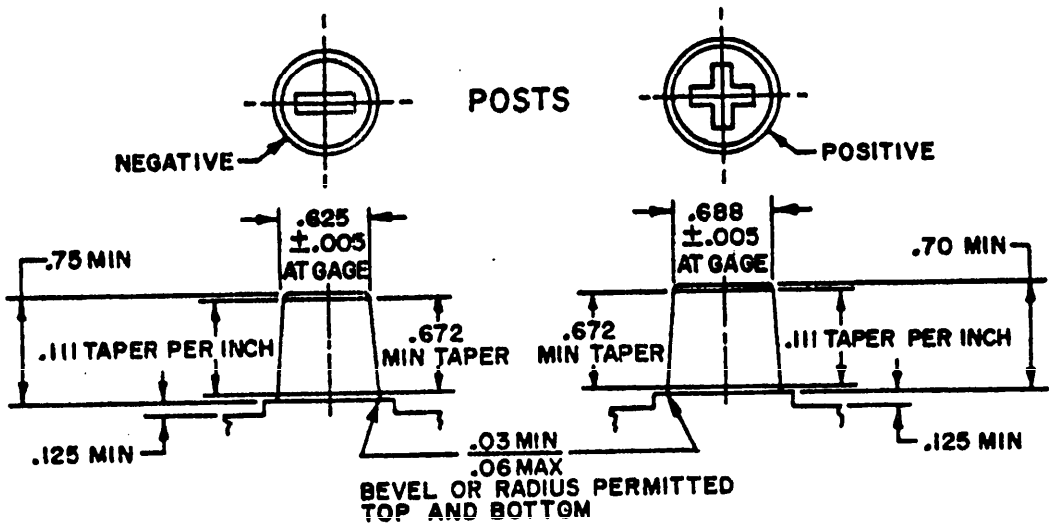
Figure 2 -BATTERY, STORAGE, NICKEL-CADMIUM

BTRY TYPE	NO OF CELLS	NOM V	AMP HOUR CAPACITY 5 HR RATE	CRANKING ABILITY		LIFE CYCLE	OVERCHARGE
				MIN TIME MINUTES	MIN 5 SEC V	MINIMUM CYCLE	MINIMUM CYCLE
				DISCH RATE 300 AMPS			
				-40°F +80°F	-40°F		
6TNC	10	12	70	5.5 10.0	8.0	300	14

NOTES:

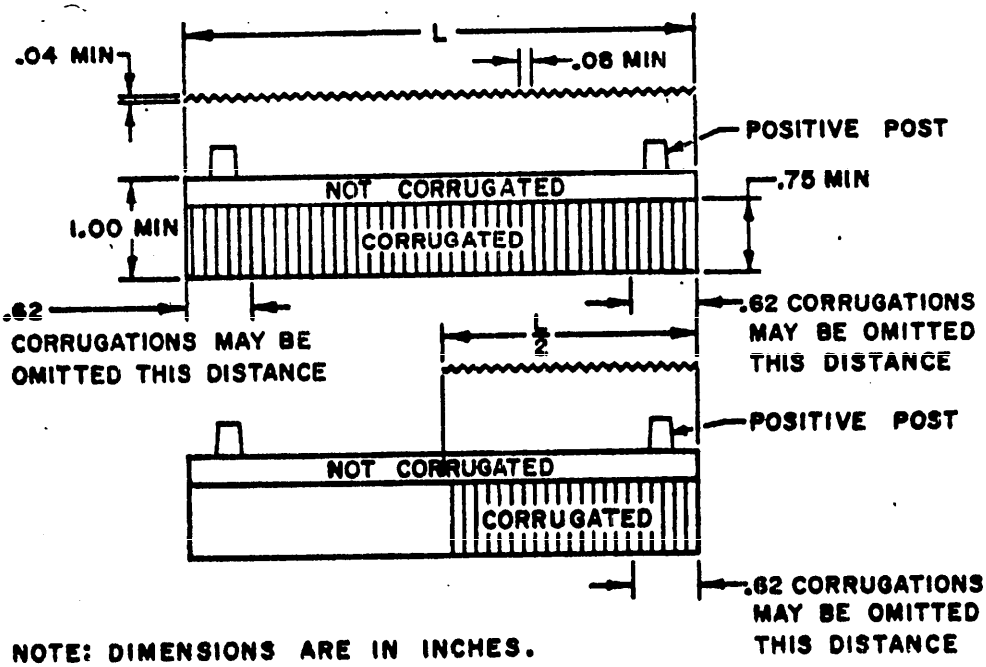
1. VENT PLUGS MUST BE REMOVABLE WITHOUT USE OF ANY TOOL OR DEVICE. PLUG LOCATIONS SHALL BE FIXED ONLY IN RESPECT TO REQUIRED MINIMUM DISTANCE FROM TERMINAL POSTS AND SIDES OF BATTERY.
2. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.

Source: https://assist.dla.mil -- Downloaded: 20260-19T16:16Z
Check the source to verify that this is the current version before use.



NOTE: DIMENSIONS ARE IN INCHES.

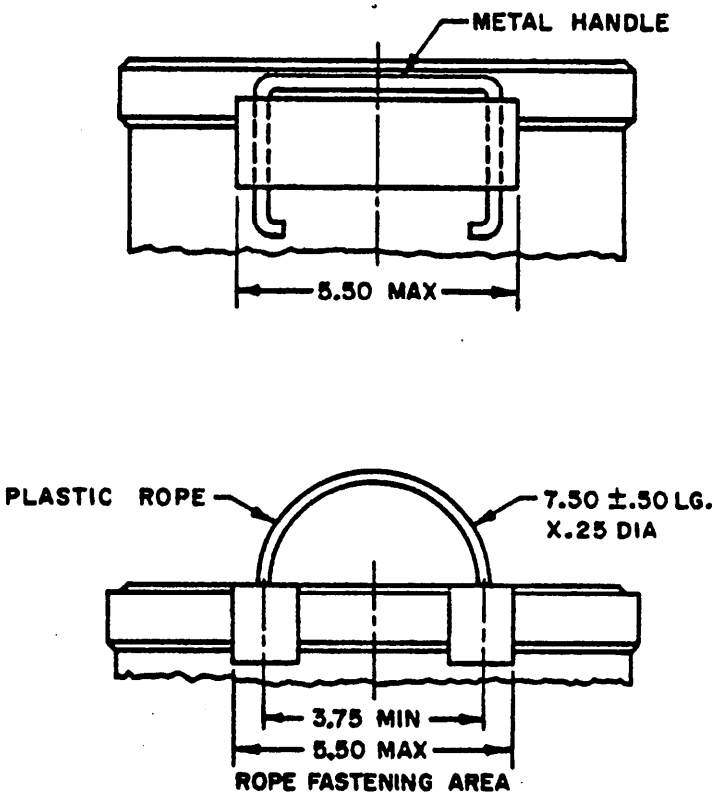
FIGURE 3 - TERMINAL POSTS



NOTE: DIMENSIONS ARE IN INCHES.

FIGURE 4 COVER CORRUGATION

Source: https://assist.dla.mil -- Downloaded: 20260-19T16:16Z
Check the source to verify that this is the current version before use.



NOTES:

1. DIMENSIONS ARE IN INCHES.
2. METAL HANDLE SHALL NOT EXTEND ABOVE TOP OF COVER WHEN IN DOWN POSITION. WHEN IN UP POSITION (LIFT) THERE SHALL BE A MINIMUM OF 1" CLEARANCE BETWEEN HANDLE AND TOP OF COVER.
3. ANCHORAGE SHALL BE BELOW BOTTOM OF COVER LEDGE.

FIGURE 5 - HANDLES
(PLASTIC ROPE CONSTRUCTION
OPTIONAL)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:16Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-B-23272B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 202610-19T16:16Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY

DEFENSE SUPPLY AGENCY
OFFICIAL BUSINESS

Commanding General
U. S. Army Tank-Automotive Command
ATTN: AMSTA-RSES
Warren, MI 48090

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:16Z
Check the source to verify that this is the current version before use.